

Shubham Khanapure

Senior ML Engineer — Generative AI & Computer Vision Specialist

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Professional Summary

Senior ML Engineer with **6+ years** building production AI/ML systems at scale. Specialized in **Generative AI** (LangChain, agentic RAG, LLMs) and **Computer Vision** (YOLO, object detection, OCR). Expert in Python, PyTorch, Azure ML, end-to-end MLOps pipelines, and distributed systems. Proven track record of reducing manual effort by 70-80% through AI automation, developing enterprise-scale GenAI applications, and delivering scalable production solutions in financial services and engineering automation domains.

Technical Skills

Languages: Python, SQL, JavaScript, Bash, Java

ML/DL Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, Hugging Face Transformers

Computer Vision: YOLO (v5/v8/v11), Faster R-CNN, Azure Custom Vision, OpenCV, SAHI, Detectron2, Image Segmentation, OCR (DocTR, Tesseract, PaddleOCR)

Generative AI & NLP: LangChain, LangGraph, LlamaIndex, GPT-4, Azure OpenAI, RAG, Guardrails AI, BERT, spaCy, NLTK

MLOps & DevOps: Azure ML Studio, MLflow, Docker, Kubernetes, CI/CD, Jenkins, Model Serving, A/B Testing, Git

Cloud & Data: Azure (ML Workspace, Databricks, Cognitive Services, Data Labeling, DevOps, Web Apps, Blob Storage, Container Registry), PostgreSQL, MySQL, MongoDB, Vector DBs (Pinecone, ChromaDB, Weaviate)

Software Engineering: RESTful APIs, Flask, FastAPI, Microservices, System Design, Agile/Scrum, Jira, Confluence, Distributed Systems

Professional Experience

Emerson Export Engineering Centre

Pune, India

Data Scientist - Generative AI & Computer Vision

April 2025 – Present

- Developed **production-grade P&ID Functional Specification Generation (FSG) system** processing high-resolution complex engineering diagrams using **YOLO object detection**, **Azure Custom Vision**, and **Azure Document Intelligence**; built end-to-end pipeline on **Azure ML Studio** with data labeling, model training via Azure Jobs, and tag text extraction using **DocTR OCR** with intelligent object-to-tag spatial mapping algorithms.
- Architected **production-level microservices** using Flask deployed on **Azure Web Apps** serving real-time predictions; integrated with **Azure ML Workspace** for model versioning and serving, implemented comprehensive experiment tracking with **MLflow**, utilized **Azure Blob Storage** for artifacts and datasets, and established automated training pipelines with Azure Jobs handling high-resolution complex P&IDs with robust preprocessing workflows.
- Built **agentic RAG system** using **LangGraph** multi-agent architecture for automated functional specification document generation; designed specialized agents for extraction, validation, and content generation with **Guardrails AI** validation ensuring compliance with engineering standards and document quality.

TEKsystems India (Client: HSBC)

Pune, India

Data Scientist - NLP & Computer Vision

July 2024 – April 2025

- Implemented **NLP and OpenCV solutions** for real-time workforce monitoring system processing employee interactions with automated activity classification and tracking; built **multi-modal OCR pipeline** using EAST/CRAFT text detection with Tesseract, PaddleOCR, and DocTR for recognition processing large-scale financial data via **PySpark on Azure Databricks**.
- Engineered **system performance monitoring infrastructure** tracking CPU, RAM, and disk utilization in UAT environment; implemented automated dashboards using Prometheus and Grafana reducing production incidents through comprehensive performance metrics, alerting, and system reliability improvements.
- Built **end-to-end CI/CD pipelines** using Azure DevOps and Docker containerization with **Azure Container Registry**; established automated testing frameworks achieving high code coverage and enabling regular production releases with comprehensive deployment automation and version control.

KPMG Global Services

Pune, India

Associate Consultant - Data Scientist

June 2022 – July 2024

- Developed and deployed **production GenAI applications** using GPT-4, Azure OpenAI, and LangChain serving enterprise users; reduced manual business process effort by **70-80%** through automated workflows including RFP response generation, document analysis, and intelligent search systems achieving significant operational efficiency improvements.

- Designed **enterprise RAG (Retrieval-Augmented Generation) architecture** with semantic search using sentence transformers and vector databases (Pinecone, Weaviate); built **full-stack AI applications** using Django and FastAPI deployed on Azure Web Apps with auto-scaling, A/B testing, and continuous integration workflows.
- Performed **LLM fine-tuning and optimization** using LoRA/QLoRA techniques improving inference efficiency; established MLOps pipelines with automated model retraining, versioning using MLflow, and monitoring for production ML systems achieving significant cost reduction while maintaining quality.
- **Awards:** Recipient of **KPMG Megastar Award** and **KPMG Encore Kudos Award** for exceptional performance and innovation in AI/ML project delivery.

Vyom Labs

Engineer - Data Scientist

Pune, India

September 2019 – May 2022

- Developed and deployed **supervised and unsupervised ML models** using PyTorch, scikit-learn, and XGBoost; built automated ticket routing system achieving **88% classification accuracy**, implemented **NLP pipelines** for text classification and NER using BERT and spaCy achieving high F1-scores, and built **automated data processing pipelines** using Apache Airflow with PCA/LDA dimensionality reduction.

Key Projects

P&ID Functional Specification Generation (FSG) — Azure ML, YOLO, Computer Vision April 2025 – Present

- Built **end-to-end production ML system** for automated functional specification generation from high-resolution P&ID engineering diagrams; implemented **object detection** using YOLO models and Azure Custom Vision with Azure ML Studio data labeling, **text extraction** using Azure Document Intelligence and DocTR OCR, and intelligent **object-to-tag spatial mapping** algorithms for accurate relationship identification.
- Deployed **Flask microservice on Azure Web Apps** integrated with Azure ML Workspace for model serving and **Azure Blob Storage** for artifact management; established complete **MLOps pipeline** with Azure Job creation for automated training, MLflow experiment tracking, model versioning, and production monitoring handling complex high-resolution engineering diagrams with robust preprocessing workflows.

Pulse Intelligence Tool — Python, Deep Learning, NLP, OpenCV, Azure, Databricks

July 2024

- Implemented **LLM-powered workforce management tool** with machine learning and automation; developed end-to-end monitoring system tracking user activities on whitelisted applications with automated pipeline from logging to report generation and visualization dashboards supporting multiple concurrent users.

CSRD and SEC Readiness Assessment Tool — Python, NLP, GenAI, Azure OpenAI

December 2023

- Developed **LLM-powered compliance assessment system** reducing manual effort by **70-80%**; implemented end-to-end pipeline from disclosure requirements to report generation with automated question generation and **RAG system** for answering with AI-powered evaluation.

Job Architecture Optimization — Python, NLP, GenAI, Prompt Engineering

December 2023

- Implemented **GenAI solution** for job architecture optimization classifying families and groups; developed decision tree for career path determination and created **Django application** deployed on Azure Web Service with production-grade scalability.

Intelligence RFP Builder — Python, NLP, Hugging Face, PyTorch, Django, Azure

January 2023

- Built **automated RFP response system** with semantic search and sentence similarity matching; extracted questions from documents, generated responses, integrated **GPT-4 chatbot**, and deployed complete Django web application on Azure using Docker.

Ticket Classification & Entity Extraction — Python, ML, NLP, Azure

January 2020

- Built machine learning model for automated ticket routing analyzing mail text and assigning to departments; achieved **88% classification accuracy** improving efficiency and reducing resolution time.

Education

Vidya Pratishthan's Kamalnayan Bajaj Institute of Engineering & Technology

Bachelor of Engineering in Computer Science

Pune, India

August 2015 – June 2019

Certifications

Professional Certifications: Azure AI Fundamentals (AI-900), Neural Networks and Deep Learning (Coursera), Data Scientist Nanodegree (Udacity), Improving Deep Neural Networks (Coursera), Machine Learning for Business Professionals (Coursera)

Awards & Recognition

KPMG Megastar Award (2024) - Exceptional performance and innovation in AI/ML projects

KPMG Encore Kudos Award (2023) - Outstanding contribution to GenAI initiatives